Non-Spacetime Attractors and the Source of Gravity

What a Relational Network Forces, What Only Relaxes, and What Must Be Declared

Matěj Rada
MatejRada@email.cz
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Abstract

The SCAR experiment removed $T_{\mu\nu}$, reconstructed curvature, and found a structured residual that does not attract. We ask the sharper question: is there an attractor, not built on spacetime, that is the *source* of gravity? We separate three meanings of "attractor": algebraic remainder, consistency selector, and dynamical attractor. We then lock what a coordinate-free relational network forces.

Locked.

(i) *Balance.* Local pairwise exchange on any network conserves the total exactly; changes are divergences of edge fluxes.

(ii) *The admissible response operator.* A nearest-neighbour linear response that is shift-blind (only differences are observable), reciprocal and isotropic is $c\times$ the graph Laplacian, with continuum limit $c\,h^2\nabla^2$.

(iii) *The constraint attractor.* On a connected network the relaxation $\dot{\phi} = -(L\phi + \kappa(\rho - \bar{\rho}))$ converges from *every* initial state to one Poisson solution, unique up to a constant (the gauge null), at the rate of the spectral gap. Five random starts reach it to 5×10^{-12}.

(iv) *The decisive test.* Same initial geometry, same rule, two sources: the attractors differ by exactly $\kappa L^{+}(\rho_A - \rho_B)$ (error 5×10^{-13}). With the coupling switched off they do not differ.

(v) *Inverse square.* In a three-dimensional network the point-source attractor is $a/r + b$ with $4\pi a = 1.055$ (finite box), i.e. a force $\propto r^{-2}$.

Refused.

- Conservative (Hamiltonian) dynamics has *no* attractor: the difference energy of two solutions is conserved (Liouville). A GR-like evolution cannot pull states to a preferred geometry. Its constraints restrict; they do not attract. SCAR's negative result is a theorem, not a failure.
- The scalar attractor is Newtonian gravity, not GR: a conformally flat (scalar) coupling bends light by $0''$, against the observed $1.751''$ (GR, $\gamma = 1$).

Bridged, with imports named. The tensor sector needs three declarations: spin-2 gauge invariance (the linearized Einstein operator annihilates pure gauge, verified symbolically), a Lorentzian network cone, and universal metric coupling. The contracted Bianchi identity (verified symbolically) then forces a conserved source, and Lovelock's theorem selects $G + \Lambda g = \kappa T$ in four dimensions. Among local diffeomorphism-invariant actions, power counting makes Einstein–Hilbert the *large-scale attractor*: higher-curvature terms scale as $(\ell k)^2 \to 0$.

Answer to "A, B or A+B". The evidence supports B: gravity's source law is a *consistency selector*. A appears only as relaxation (descriptive or dissipative), never as conservative evolution. The non-spacetime attractor that exists is the Poisson constraint on the network. Its tensor completion is selected, not attracted.

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1 Three attractors, kept apart

Meaning	Statement	What proves it
Algebraic remainder	something survives a subtraction (SCAR)	an exact decomposition
Consistency selector	only configurations satisfying $C_i = 0$ are admissible	existence and uniqueness of solutions to $C_i = 0$
Dynamical attractor	an evolution drives a basin to a set	a law, a basin, stability, convergence

There is also a fourth, *coarse-graining attraction*: a flow in descriptive scale toward a stable effective law (Proposition 5.4). It is not a flow in physical time.

No-Go 1.1 (Conservative dynamics has no attractor). For a linear conservative network dynamics $\ddot{u} = -Lu$, the energy $E[u] = \frac{1}{2}\|\dot{u}\|^2 + \frac{1}{2}u^\top Lu$ is conserved, and so is $E[u_A - u_B]$ for any two solutions. More generally, Hamiltonian flows preserve phase-space volume (Liouville), so no open set contracts onto an attractor.

Proof. $\frac{d}{dt}E = \dot{u}^\top\ddot{u} + \dot{u}^\top Lu = 0$. Linearity transfers this to differences. Witness (G9): a symplectic leapfrog on a 300-node network keeps the energy within 0.9%, and the difference energy moves from 0.06604 to 0.06628, with no contraction. □

Corollary 1.2 (What SCAR's negative result means)**.** *GR's evolution is Hamiltonian (ADM), with first-class constraints. No residual of a GR reconstruction can be a dynamical attractor of GR's own evolution. SCAR is at most a* constraint residue*: a diagnostic of what the observable geometry does not determine. The search must move to consistency selectors.*

2 The relational network, without spacetime

Assumption N1 (Network)**.** A finite connected graph (V, E). Nodes carry operational loads ρ_v and potentials ϕ_v. Edges carry exchanges. There are no coordinates, distances or metric.

Assumption N2 (Admissible response)**.** The response of ϕ to ρ is linear and nearest-neighbour, and it is:

- *shift-blind:* only differences of ϕ are observable, so adding a constant changes nothing;
- *reciprocal:* the influence of v on w equals that of w on v;
- *isotropic:* on a lattice, invariant under the lattice symmetry.

Proposition 2.1 (Balance; G1)**.** *Let B be the signed incidence matrix. Exchange $\dot{\rho} = -B^\top f$ for any edge flux f conserves $\sum_v \rho_v$ exactly, because each row of B sums to zero, so $\mathbf{1}^\top B^\top f = 0$: each edge removes from one node what it adds to the other. This is the discrete balance law $\nabla \cdot J = -\partial_t \rho$, derived without a metric.*

Proposition 2.2 (The admissible operator is the Laplacian; G2)**.** *Under Assumption N2 on the square lattice, the stencil has centre weight $-4a_1$ and equal neighbour weights a_1, i.e. it is $-a_1 L$. Its Taylor expansion is $a_1 h^2 \nabla^2 f + O(h^4)$. On a general network, reciprocity makes the response symmetric, and shift-blindness makes its rows sum to zero, so it is a weighted graph Laplacian $L = B^\top W B$.*

Isotropy fixes equal weights only where the network has a symmetry. On an irregular network the weights are data, and they are exactly the place a metric enters (Section 5).

3 The constraint attractor

Theorem 3.1 (Constraint attractor)**.** *Under Assumptions N1–N2, consider the relaxation*

$$\dot{\phi} = -\big(L\phi + \kappa(\rho - \bar{\rho}\mathbf{1})\big).$$

(a) Every solution converges, modulo constants, to the unique zero-mean solution ϕ^* of the Poisson constraint $L\phi^* = -\kappa(\rho - \bar{\rho}\mathbf{1})$.

(b) The convergence is exponential at the rate of the spectral gap $\lambda_2(L)$.

(c) The constant mode is exactly neutral: it is the gauge null of Assumption N2.

(d) Source differences map linearly: $\phi^*_A - \phi^*_B = -\kappa L^+(\rho_A - \rho_B)$, and with $\kappa = 0$ the attractors coincide.

Proof. L is symmetric positive semidefinite with kernel spanned by $\mathbf{1}$ on a connected graph. The source term is orthogonal to $\mathbf{1}$, so the Poisson equation is solvable on $\mathbf{1}^\perp$. Write $\phi = c\mathbf{1} + \psi$ with $\psi \perp \mathbf{1}$. Then $\dot{c} = 0$ and $\dot{\psi} = -L(\psi - \phi^*)$, so $\|\psi - \phi^*\|$ decays as $e^{-\lambda_2 t}$. Part (d) is linearity of L^+. □

Witnesses (G3, G4). On a 300-node seeded random network with spectral gap 0.448:

- five random initial states converge to ϕ^* within 5.0×10^{-12};
- two sources under the same rule differ at the attractor by $\kappa L^+(\rho_A - \rho_B)$ within 4.8×10^{-13};
- with the coupling off, the difference is exactly 0.

Proposition 3.2 (Inverse square; G5)**.** *On a three-dimensional cubic network (41^3 nodes, Dirichlet boundary), the response to a point load is $a/r + b$ on $r = 3, \ldots, 9$, with $4\pi a = 1.055$ and relative fit error 0.56%. The excess over 1 is the finite-box and lattice correction. The force is therefore $\propto r^{-2}$. On a network of spectral dimension d the decay is r^{2-d}, so the exponent records the network's dimension. It does not fix it.*

Reading. This is a genuine attractor not built on spacetime. It acts on a relational network, it is selected by balance and admissible response alone, and it converges from every state. Its content is Newtonian gravity: $\nabla^2\phi = 4\pi G\rho$, with G the single anchor κ.

4 Why the scalar attractor is not the source of GR

No-Go 4.1 (A scalar cannot bend light correctly; G6). Any scalar coupling makes the metric conformally flat, $g = (1 + 2\phi)\eta$. Null curves are conformally invariant, so light is not deflected: $0''$ at the solar limb, i.e. $\gamma = -1$. Observation gives $1.751''$, i.e. $\gamma = 1$. The Cassini bound is $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [1].

The scalar attractor reproduces Newton exactly and fails at the first relativistic test. The missing part is the *tensor* sector, i.e. spatial curvature, which is what γ measures.

5 The tensor completion: selected, not attracted

Assumption N3 (Lorentzian cone)**.** Influence on the network propagates inside one declared Lorentzian characteristic cone, the same for all matter (Monotonicity and the Lorentz Sign; RG Gravitation).

Assumption N4 (Universal coupling)**.** All matter couples to one symmetric tensor field $h_{\mu\nu}$, and only through it: the equivalence principle, declared.

Assumption N5 (Gauge redundancy)**.** The physical content of h is invariant under $h \mapsto h + \partial\xi + \partial\xi$: the tensor analogue of shift-blindness.

Proposition 5.1 (Gauge invariance forces conservation; G10)**.** *The linearized Ricci operator annihilates every pure-gauge $h_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$. This is verified symbolically on a plane-wave fixture with arbitrary profile functions. A source law $\mathcal{E}[h] = \kappa T$ with gauge-invariant $\mathcal{E}$ therefore requires $\partial^\mu T_{\mu\nu} = 0$.*

Proposition 5.2 (Nonlinear consistency; G8)**.** *For a general static spherical metric, $\nabla_\mu G^{\mu\nu} = 0$ identically (symbolic). The source law $G = \kappa T$ is consistent exactly with covariant conservation $\nabla_\mu T^{\mu\nu} = 0$, the continuum form of the balance law of Section 2 on a curved background.*

Imported Theorem 5.3 (Selection [2, 3])**.** *Self-consistent coupling of a massless spin-2 field to its own conserved stress energy reconstructs the Einstein equation (Deser). Lovelock's theorem: in four dimensions, the only divergence-free symmetric tensor built from the metric and its first two derivatives is $aG_{\mu\nu} + \Lambda g_{\mu\nu}$.*

Proposition 5.4 (The scale attractor; G7)**.** *For a weak wave of wavenumber k, an R^2 term contributes $\ell^2 k^4 h_k^2$, against $k^2 h_k^2$ for Einstein–Hilbert: a ratio $(\ell k)^2 \to 0$ as $k \to 0$. Among local diffeomorphism-invariant actions, the lowest-derivative terms dominate at large scales. Einstein–Hilbert plus Λ is the infrared attractor of the coarse-graining flow, while higher-curvature terms are irrelevant directions.*

This is the coarse-graining attraction of Section 1. It explains why the macroscopic law is robust to microscopic detail. It does not make individual solutions converge.

6 The six obligations, typed

Obligation	What is locked	Status
Operational state reconstruction	predictive quotient (From the Wall-Tap); static spherical metric from clock ratios and differential radar, $B = \frac{c}{2} z T'$ (round-trip report)	LOCKED, conditional on anchors
Independent source characterization	discrete balance (G1); stress as second kinetic moment (Stress Tensor paper)	balance FORCED; tensor form needs N3
Dynamical admissibility principle	shift-blind reciprocal isotropic response = Laplacian (G2); tensor gauge invariance (G10)	scalar FORCED; tensor DECLARED (N5)
Geometric source–response relation	scalar: Poisson attractor (G3–G5); tensor: $G + \Lambda g = \kappa T$	scalar FORCED; tensor SELECTED (Lovelock, imported)
Consistency and evolution	Bianchi $\Leftrightarrow$ conservation (G8); conservative evolution has no attractor (G9)	LOCKED
Uniqueness and prediction	κ fixed once by the Newtonian attractor; then deflection $1.751''$, Mercury $42.98''$/cy, GPS and Pound–Rebka shifts (RG Gravitation)	CONDITIONAL on N3–N5 and $D = 4$

7 What remains declared

- **Dimension.** The decay exponent records the network's spectral dimension. It does not derive $D = 4$.
- **The cone.** Lorentzian propagation needs one declared non-monotone direction (Monotonicity and the Lorentz Sign).
- **Universality of free fall.** One metric for all matter is declared (N4). The network does not yet show why different internal compositions average to one effective geometry. That is the open *universality theorem.*
- **Tensor gauge invariance.** Shift-blindness forces the scalar Laplacian; the spin-2 analogue (N5) is a declaration, not yet derived from network rules.
- G **and** Λ**.** One anchor each.
- **Nonlinear completion.** Imported (Deser, Lovelock); only cubic consistency has been checked in RG's batteries.

The theorem we can state. *On any connected relational network, balance and admissible (shift-blind, reciprocal) response force a Poisson constraint, and its relaxation is a global attractor modulo gauge whose response to source changes is linear and uniquely determined. Under a declared Lorentz cone, universal tensor coupling and tensor gauge invariance, the consistent source law in four dimensions is Einstein's, and it is the large-scale attractor of the local action space.*

Outcome classification. In the terms of the Independent Source–Geometry Emergence programme:

- *Unique* for the scalar (Newtonian) sector, from network axioms alone.
- *Unique under declarations* for the tensor sector.
- *Failure* of every attempt to obtain GR as a dynamical attractor in physical time (No-Go 1.1).

8 Reproduction and limits

- `rg_gravity_source.py` runs G1–G10 with fixed seeds. G2, G6–G8 and G10 are symbolic. Reruns are byte-identical.
- *Limits.*
  - One random network topology and one cubic lattice.
  - The relaxation is a declared dissipative dynamics. It models a selector, not physical time evolution.
  - The Green's-function fit includes box and lattice corrections ($4\pi a = 1.055$).
  - The tensor sector is imported beyond linear gauge invariance and spherical Bianchi checks.
  - No real data are fitted here.

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